Problem 1: Smart Home Energy Consumption (Bayesian Network)

Team GigaChad: Siddhi Singh (B24DS508), Anushka Gawande (B24CS007), Anu Yadav (B24EE007), Anjil Gupta (B24EE004), Bhumi Tapdiya (B24EE011)

Part A: Inference (Expert Model)

Query	Value
Q1: P(E=High)	0.5806
Q2: P(E=High | W=Hot)	0.7063
Q3: P(E=High | D=Weekend)	0.6402
Q4: P(E=High | W=Cold, O=High)	0.4843
Q5: P(W=Hot | E=High)	0.6082
Q6: P(AC=On | E=High)	0.6929
Q7: P(AC=On | E=High, L=Heavy)	0.6499

Q6 vs Q7: Explaining away

AC and Appliance Load are both causes of high Energy. Knowing E=High raises the belief that AC is on. When we also learn that the load is Heavy, the heavy load already explains the high consumption, so the AC is less needed as an explanation. P(AC=On) therefore drops from 0.6929 to 0.6499.

Q1 by hand

P(E=High) = Σ P(w) P(d) P(o|d) P(a|w,o) P(l|o) P(E=High|a,l)

1. P(O=High) = 0.7(0.45) + 0.3(0.80) = 0.555, so P(O=Low) = 0.445
2. P(A=On | O=High) = 0.5(0.9) + 0.3(0.4) + 0.2(0.1) = 0.59
3. P(A=On | O=Low) = 0.5(0.6) + 0.3(0.15) + 0.2(0.05) = 0.355
4. P(E=High | O=High) = 0.59(0.75)(0.92) + 0.59(0.25)(0.68) + 0.41(0.75)(0.55) + 0.41(0.25)(0.12) = 0.688825
5. P(E=High | O=Low) = 0.355(0.35)(0.92) + 0.355(0.65)(0.68) + 0.645(0.35)(0.55) + 0.645(0.65)(0.12) = 0.445693
6. P(E=High) = 0.555(0.688825) + 0.445(0.445693) = 0.580631

This matches the value from the code (0.580631).

Part B: Conditional Independence

Statement	Holds?	Reason (d-separation)	Numerical check
W ⊥ D	Yes	Paths W → AC ← O ← D and W → AC → E ← L ← O ← D are blocked at the unobserved colliders AC and E.	P(W=Hot) = 0.5000, P(W=Hot | D=Weekend) = 0.5000
W ⊥ D | E	No	Observing E opens collider E, and also collider AC (E is its descendant). Both paths become active.	P(W=Hot | E=High) = 0.6082, P(W=Hot | E=High, D=Weekend) = 0.5998
W ⊥ L | O	Yes	Path W → AC ← O → L is blocked at observed O (and at collider AC). Path W → AC → E ← L is blocked at unobserved collider E.	P(W=Hot | O=High) = 0.5000, P(W=Hot | O=High, L=Heavy) = 0.5000

Part C: Synthetic Data

Ancestral sampling in topological order (Weather, DayType, Occupancy, AC, ApplianceLoad, Energy). Random seed = 42.

Weather frequency:

n	Hot	Mild	Cold
500	0.510	0.286	0.204
5000	0.4992	0.3020	0.1988
20000	0.4953	0.3058	0.1989

(Weather, Occupancy) counts:

n	Hot,High	Hot,Low	Mild,High	Mild,Low	Cold,High	Cold,Low
500	155	100	75	68	51	51
5000	1402	1094	823	687	558	436
20000	5520	4386	3416	2700	2228	1750

Part D: Learned AC CPT, P(AC=On | W, O)

W, O	Expert	MLE 500	Bayes 500	MLE 5000	Bayes 5000	MLE 20000	Bayes 20000
Hot, High	0.90	0.8516	0.8479	0.8937	0.8933	0.9013	0.9011
Hot, Low	0.60	0.6600	0.6574	0.6188	0.6186	0.6008	0.6007
Mild, High	0.40	0.3600	0.3630	0.3900	0.3903	0.3970	0.3970
Mild, Low	0.15	0.1765	0.1842	0.1514	0.1522	0.1478	0.1480
Cold, High	0.10	0.0980	0.1108	0.1039	0.1051	0.1041	0.1044
Cold, Low	0.05	0.0196	0.0348	0.0482	0.0499	0.0537	0.0541

Part E: Comparative Analysis

n = 500:

Query	Expert	MLE	Bayesian	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5570	0.5559	0.0237	0.0247
Q2	0.7063	0.7050	0.6991	0.0013	0.0072
Q3	0.6402	0.6092	0.6046	0.0310	0.0356
Q4	0.4843	0.4203	0.4268	0.0639	0.0575
Q5	0.6082	0.6455	0.6370	0.0373	0.0288
Q6	0.6929	0.7280	0.7236	0.0351	0.0307
Q7	0.6499	0.6584	0.6560	0.0085	0.0061

n = 5000:

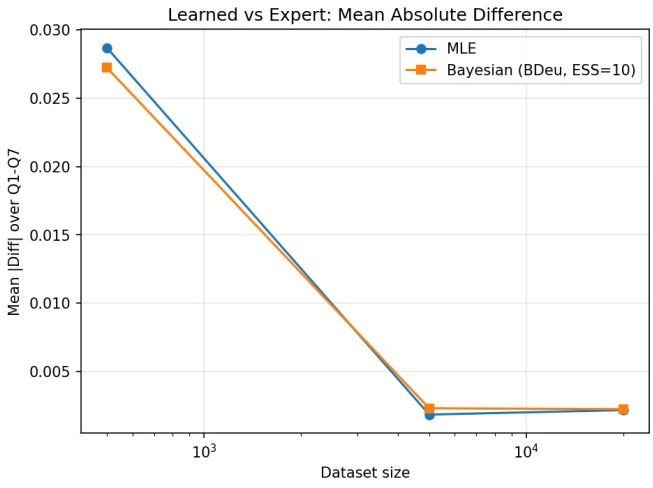
Query	Expert	MLE	Bayesian	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5794	0.5793	0.0012	0.0014
Q2	0.7063	0.7034	0.7028	0.0029	0.0035
Q3	0.6402	0.6399	0.6393	0.0003	0.0009
Q4	0.4843	0.4860	0.4863	0.0017	0.0021
Q5	0.6082	0.6060	0.6053	0.0022	0.0029
Q6	0.6929	0.6884	0.6881	0.0044	0.0048
Q7	0.6499	0.6496	0.6493	0.0003	0.0006

n = 20000:

Query	Expert	MLE	Bayesian	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5768	0.5768	0.0038	0.0038
Q2	0.7063	0.7028	0.7026	0.0035	0.0036
Q3	0.6402	0.6381	0.6379	0.0021	0.0023
Q4	0.4843	0.4838	0.4839	0.0004	0.0004
Q5	0.6082	0.6035	0.6033	0.0047	0.0049
Q6	0.6929	0.6922	0.6921	0.0007	0.0008
Q7	0.6499	0.6499	0.6499	0.0001	0.0000

1. Mean absolute difference vs dataset size

n	MLE	Bayesian
500	0.0287	0.0272
5000	0.0019	0.0023
20000	0.0022	0.0023



2. Largest error at n = 500

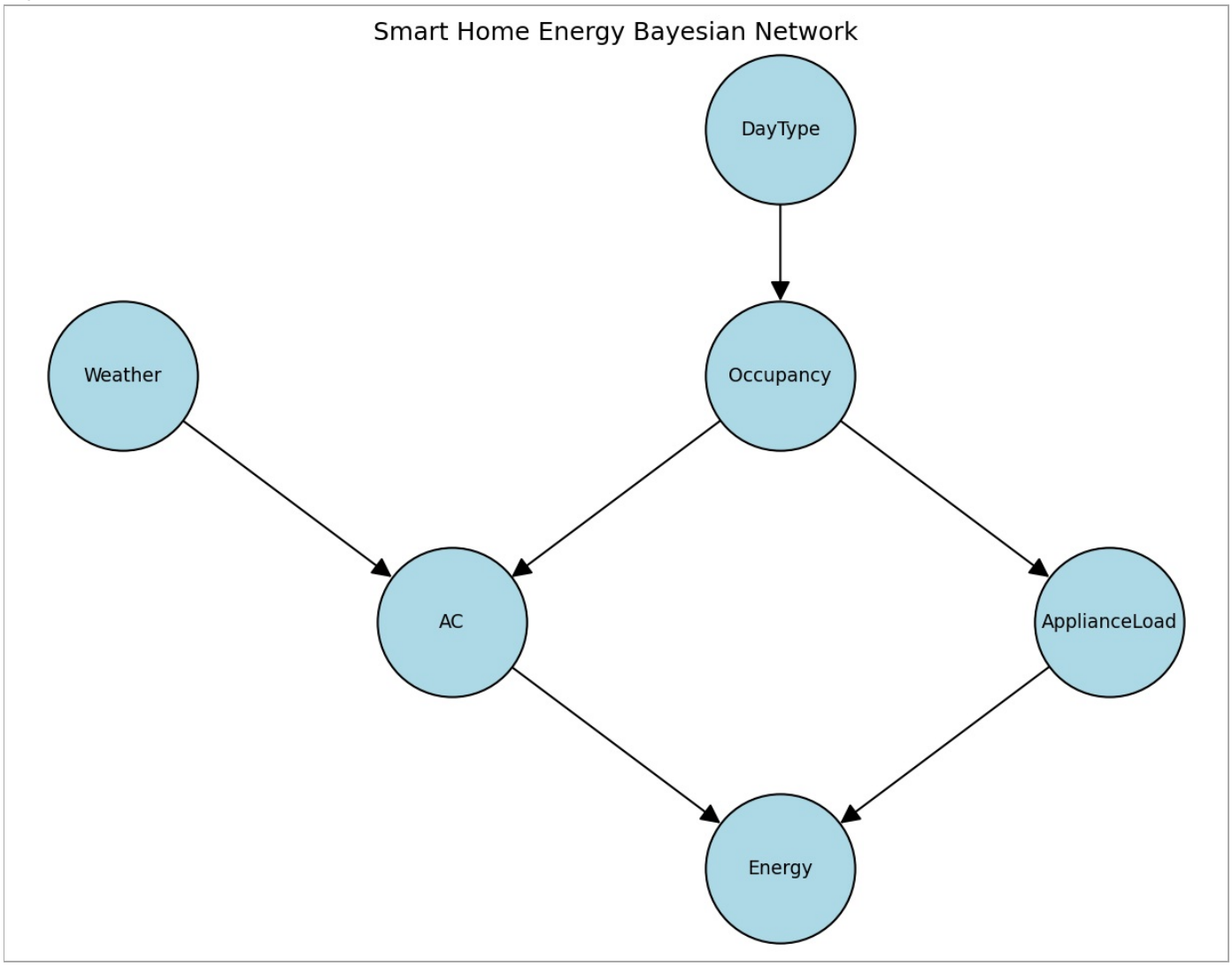
Q4 has the largest error (0.0639 for MLE, 0.0575 for Bayesian). Its evidence (Weather=Cold, Occupancy=High) is the rarest, occurring in only 51 of 500 samples. P(AC | Cold, High) is estimated from very little data, so it is the least accurate.

3. Which estimator is better for small datasets?

Bayesian estimation performs better on small data (mean error 0.0272 vs 0.0287 at n = 500). The BDeu prior adds pseudo-counts to every parent combination, which smooths estimates based on few samples and avoids extreme values. For example, P(AC=On | Cold, Low) is 0.0196 with MLE but 0.0348 with Bayesian, while the true value is 0.05. MLE uses only raw counts, so it overfits the noise in small samples. As the dataset grows, the prior's effect fades and both estimators give nearly the same results.

Screenshots

Bayesian Network structure



Part A: CPT verification and CPTs

```
$ python smart_home_bn.py

Part A: CPT verification
Weather        column sums=[1.] -> OK
DayType        column sums=[1.] -> OK
Occupancy      column sums=[1. 1.] -> OK
AC             column sums=[1. 1. 1. 1. 1. 1.] -> OK
ApplianceLoad  column sums=[1. 1.] -> OK
Energy         column sums=[1. 1. 1. 1.] -> OK
model.check_model(): True
+---------------+-----+
| Weather(Hot)  | 0.5 |
+---------------+-----+
| Weather(Mild) | 0.3 |
+---------------+-----+
| Weather(Cold) | 0.2 |
+---------------+-----+
+------------------+-----+
| DayType(Weekday) | 0.7 |
+------------------+-----+
| DayType(Weekend) | 0.3 |
+------------------+-----+
+-----------------+------------------+------------------+
| DayType         | DayType(Weekday) | DayType(Weekend) |
+-----------------+------------------+------------------+
| Occupancy(High) | 0.45             | 0.8              |
+-----------------+------------------+------------------+
| Occupancy(Low)  | 0.55             | 0.2              |
+-----------------+------------------+------------------+
+-----------+---------------------+----------------+-----------------+----------------+-----------------+----------------+
| Weather   | Weather(Hot)        | Weather(Hot)   | Weather(Mild)   | Weather(Mild)  | Weather(Cold)   | Weather(Cold)  |
+-----------+---------------------+----------------+-----------------+----------------+-----------------+----------------+
| Occupancy | Occupancy(High)     | Occupancy(Low) | Occupancy(High) | Occupancy(Low) | Occupancy(High) | Occupancy(Low) |
+-----------+---------------------+----------------+-----------------+----------------+-----------------+----------------+
| AC(On)    | 0.9                 | 0.6            | 0.4             | 0.15           | 0.1             | 0.05           |
+-----------+---------------------+----------------+-----------------+----------------+-----------------+----------------+
| AC(Off)   | 0.09999999999999998 | 0.4            | 0.6             | 0.85           | 0.9             | 0.95           |
+-----------+---------------------+----------------+-----------------+----------------+-----------------+----------------+
+----------------------+-----------------+----------------+
| Occupancy            | Occupancy(High) | Occupancy(Low) |
+----------------------+-----------------+----------------+
| ApplianceLoad(Heavy) | 0.75            | 0.35           |
+----------------------+-----------------+----------------+
| ApplianceLoad(Light) | 0.25            | 0.65           |
+----------------------+-----------------+----------------+
+---------------+----------------------+----------------------+----------------------+----------------------+
| AC            | AC(On)               | AC(On)               | AC(Off)              | AC(Off)              |
+---------------+----------------------+----------------------+----------------------+----------------------+
| ApplianceLoad | ApplianceLoad(Heavy) | ApplianceLoad(Light) | ApplianceLoad(Heavy) | ApplianceLoad(Light) |
+---------------+----------------------+----------------------+----------------------+----------------------+
| Energy(High)  | 0.92                 | 0.68                 | 0.55                 | 0.12                 |
+---------------+----------------------+----------------------+----------------------+----------------------+
| Energy(Low)   | 0.07999999999999996  | 0.31999999999999995  | 0.44999999999999996  | 0.88                 |
+---------------+----------------------+----------------------+----------------------+----------------------+
saved bn_structure.png
```

Part A: queries and hand calculation, Part B

```
$ python smart_home_bn.py   (continued)

Part A: expert queries
Q1: 0.580631
Q2: 0.706266
Q3: 0.640199
Q4: 0.484250
Q5: 0.608188
Q6: 0.692867
Q7: 0.649899

Hand calculation of Q1
P(O=High) = 0.7*0.45 + 0.3*0.80 = 0.5550;  P(O=Low) = 0.4450
P(A=On|O=High) = 0.5*0.9 + 0.3*0.4 + 0.2*0.1 = 0.5900
P(A=On|O=Low)  = 0.5*0.6 + 0.3*0.15 + 0.2*0.05 = 0.3550
P(E=High|O=High) = 0.590*0.75*0.92 + 0.590*0.25*0.68 + 0.410*0.75*0.55 + 0.410*0.25*0.12 = 0.688825
P(E=High|O=Low) = 0.355*0.35*0.92 + 0.355*0.65*0.68 + 0.645*0.35*0.55 + 0.645*0.65*0.12 = 0.445693
P(E=High) = 0.555*P(E=High|O=High) + 0.445*P(E=High|O=Low) = 0.580631
Code Q1 = 0.580631, hand Q1 = 0.580631, match = True

Part B: numerical check
1) P(W=Hot)=0.500000  P(W=Hot|D=Weekend)=0.500000  -> independent
2) P(W=Hot|E=High)=0.608188  P(W=Hot|E=High,D=Weekend)=0.599751  -> dependent
3) P(W=Hot|O=High)=0.500000  P(W=Hot|O=High,L=Heavy)=0.500000  -> independent
```

Part C: synthetic data

```
$ python smart_home_bn.py   (continued)

Part C: synthetic data (seed = 42 )
Topological order: ['Weather', 'DayType', 'Occupancy', 'AC', 'ApplianceLoad', 'Energy']

--- n = 500 ---
Weather frequency:
Weather
Hot     0.510
Mild    0.286
Cold    0.204
(Weather, Occupancy) counts:
Hot   High    155
      Low     100
Mild  High     75
      Low      68
Cold  High     51
      Low      51

--- n = 5000 ---
Weather frequency:
Weather
Hot     0.4992
Mild    0.3020
Cold    0.1988
(Weather, Occupancy) counts:
Hot   High    1402
      Low     1094
Mild  High     823
      Low      687
Cold  High     558
      Low      436

--- n = 20000 ---
Weather frequency:
Weather
Hot     0.4953
Mild    0.3058
Cold    0.1989
(Weather, Occupancy) counts:
Hot   High    5520
      Low     4386
Mild  High    3416
      Low     2700
Cold  High    2228
      Low     1750
```

Parts D and E: n = 500

```
$ python smart_home_bn.py   (continued)

===== Part D: n = 500 =====
P(AC=On | Weather, Occupancy):
                  Expert     MLE  Bayesian
Weather Occupancy
Hot     High         0.90  0.8516    0.8479
        Low          0.60  0.6600    0.6574
Mild    High         0.40  0.3600    0.3630
        Low          0.15  0.1765    0.1842
Cold    High         0.10  0.0980    0.1108
        Low          0.05  0.0196    0.0348

===== Part E: comparison, n = 500 =====
       Expert Model  MLE Model  Bayesian Model  |Diff| MLE  |Diff| Bayesian
Query
Q1           0.5806     0.5570          0.5559      0.0237           0.0247
Q2           0.7063     0.7050          0.6991      0.0013           0.0072
Q3           0.6402     0.6092          0.6046      0.0310           0.0356
Q4           0.4843     0.4203          0.4268      0.0639           0.0575
Q5           0.6082     0.6455          0.6370      0.0373           0.0288
Q6           0.6929     0.7280          0.7236      0.0351           0.0307
Q7           0.6499     0.6584          0.6560      0.0085           0.0061
```

Parts D and E: n = 5000

```
$ python smart_home_bn.py   (continued)

===== Part D: n = 5000 =====
P(AC=On | Weather, Occupancy):
                  Expert     MLE  Bayesian
Weather Occupancy
Hot     High         0.90  0.8937    0.8933
        Low          0.60  0.6188    0.6186
Mild    High         0.40  0.3900    0.3903
        Low          0.15  0.1514    0.1522
Cold    High         0.10  0.1039    0.1051
        Low          0.05  0.0482    0.0499

===== Part E: comparison, n = 5000 =====
       Expert Model  MLE Model  Bayesian Model  |Diff| MLE  |Diff| Bayesian
Query
Q1           0.5806     0.5794          0.5793      0.0012           0.0014
Q2           0.7063     0.7034          0.7028      0.0029           0.0035
Q3           0.6402     0.6399          0.6393      0.0003           0.0009
Q4           0.4843     0.4860          0.4863      0.0017           0.0021
Q5           0.6082     0.6060          0.6053      0.0022           0.0029
Q6           0.6929     0.6884          0.6881      0.0044           0.0048
Q7           0.6499     0.6496          0.6493      0.0003           0.0006
```

Parts D and E: n = 20000 and summary

```
$ python smart_home_bn.py   (continued)

===== Part D: n = 20000 =====
P(AC=On | Weather, Occupancy):
                  Expert     MLE  Bayesian
Weather Occupancy
Hot     High         0.90  0.9013    0.9011
        Low          0.60  0.6008    0.6007
Mild    High         0.40  0.3970    0.3970
        Low          0.15  0.1478    0.1480
Cold    High         0.10  0.1041    0.1044
        Low          0.05  0.0537    0.0541

===== Part E: comparison, n = 20000 =====
       Expert Model  MLE Model  Bayesian Model  |Diff| MLE  |Diff| Bayesian
Query
Q1           0.5806     0.5768          0.5768      0.0038           0.0038
Q2           0.7063     0.7028          0.7026      0.0035           0.0036
Q3           0.6402     0.6381          0.6379      0.0021           0.0023
Q4           0.4843     0.4838          0.4839      0.0004           0.0004
Q5           0.6082     0.6035          0.6033      0.0047           0.0049
Q6           0.6929     0.6922          0.6921      0.0007           0.0008
Q7           0.6499     0.6499          0.6499      0.0001           0.0000

Mean |Diff| over Q1-Q7:
           MLE  Bayesian
500    0.02869   0.02723
5000   0.00186   0.00231
20000  0.00219   0.00226
saved mean_abs_diff.png

Largest error at n=500: Q4
  Q1: evidence rows in data = 500
  Q2: evidence rows in data = 255
  Q3: evidence rows in data = 157
  Q4: evidence rows in data = 51
  Q5: evidence rows in data = 280
  Q6: evidence rows in data = 280
  Q7: evidence rows in data = 198
```